

Failure of the proposed endpoint Hölder inequality

A counterexample to arXiv:1906.10772, Open Question (ii)

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Status. High-confidence candidate counterexample, pending human review. The construction gives a complete negative answer to Open Question (ii) as literally stated; no claim is made about questions (i) or (iii).

Source conjecture

Miana and Oliva-Maza, *Generalized Stieltjes and other integral operators on Sobolev–Lebesgue spaces*, arXiv:1906.10772, conjecture that whenever $1/r = 1/p + 1/q$ with $p, q, r \in [1, \infty]$, pointwise multiplication satisfies

$$\|fg\|_{\alpha, r} \leq C_\alpha \|f\|_{\alpha, p} \|g\|_{\alpha, q}. \quad (\text{H})$$

Crucially, the displayed range includes $p = \infty$. Their Remark 3.1 proposes

$$\|f\|_{\alpha, \infty} = \frac{1}{\Gamma(\alpha + 1)} \operatorname{ess\,sup}_{t > 0} t^\alpha |W_+^\alpha f(t)|,$$

and explicitly observes that this need not control $\|f\|_\infty$.

$$\mathcal{S}_{\beta, \mu} F(z) := z^{\mu - \beta} \int_0^\infty \frac{s^{\beta - 1}}{(z + s)^\mu} F(s) ds, \quad z \in \mathbb{C}^+,$$

on the spaces $H_p^{(n)}$ for $0 < \beta - 1/p < \mu$. This will be the focus of a forthcoming paper.

8.3 Open questions

In this paper we have presented a complete study of generalized Stieltjes operator on fractional Sobolev–Lebesgue spaces. However some of these results might be improved.

- (i) The case $p = \infty$ in the Sobolev–Lebesgue spaces $\mathcal{T}_\infty^{(\alpha)}(t^\alpha)$ is commented in Remark 3.1. It seems natural to conjecture that the operator $\mathcal{S}_{\beta, \mu}$ is bounded on these spaces $\mathcal{T}_\infty^{(\alpha)}(t^\alpha)$, $\|\mathcal{S}_{\beta, \mu}\| = B(\beta, \mu - \beta)$, for $0 < \beta < \mu$ and $\sigma(\mathcal{S}_{\beta, \mu}) = \Upsilon_{\beta, \mu} \cup \{0\}$.
- (ii) In Theorem 3.2 we have proved a Hölder inequality in $\mathcal{T}_p^{(\alpha)}(t^\alpha)$ for conjugate exponents $p, p' > 1$ and $\alpha \geq 0$. As in the classical case, we may conjecture that given $f \in \mathcal{T}_p^{(\alpha)}(t^\alpha)$ and $g \in \mathcal{T}_q^{(\alpha)}(t^\alpha)$ then $fg \in \mathcal{T}_r^{(\alpha)}(t^\alpha)$ and

$$\|fg\|_{\alpha, r} \leq C_\alpha \|f\|_{\alpha, p} \|g\|_{\alpha, q},$$

where C_α is a positive constant and $r, p, q \in [1, \infty]$ such that $\frac{1}{r} = \frac{1}{p} + \frac{1}{q}$. Note that for $\alpha \in \mathbb{N}$, the statement holds in a straightforward way from the usual Leibniz formula and Hölder inequality.

The counterexample

At $\alpha = 1$, one has $W_+^1 f = -f'$ and $\Gamma(2) = 1$, hence

$$\|f\|_{1,s} = \|-tf'(t)\|_{L^s(0,\infty)} \quad \text{quad} (1 \leq s < \infty), \quad \|f\|_{1,\infty} = \operatorname{ess\,sup}_{t>0} t|f'(t)|.$$

For finite s , the source proves that $D_+^1 f = -tf'$ is an isometry from $\mathcal{T}_s^{(1)}(t)$ onto $L^s(0, \infty)$, with inverse

$$((D_+^1)^{-1}h)(t) = \int_t^\infty \frac{h(s)}{s} ds. \quad (1)$$

Theorem 1. *For every $q \in (1, \infty)$, multiplication does not map*

$$\mathcal{T}_\infty^{(1)}(t) \times \mathcal{T}_q^{(1)}(t) \longrightarrow \mathcal{T}_q^{(1)}(t)$$

under either the maximal finite-norm interpretation of the first space or the natural completion of the Schwartz core. Consequently (H) is false with $\alpha = 1$, $p = \infty$, and $r = q$.

Proof. Choose $a \in (0, 1)$ and put $b = a + 1/q$. Let $\eta \in C_c^\infty([0, \infty))$ equal 1 on $(0, 1/4)$ and 0 on $(1/2, \infty)$. With $L(t) = \log(e/t)$, define for $t > 0$

$$f(t) = \eta(t)L(t)^a, \quad g(t) = \eta(t)t^{-1/q}L(t)^{-b}.$$

On $(0, 1/4)$,

$$-tf'(t) = aL(t)^{a-1} \longrightarrow 0, \quad -tg'(t) = \left(\frac{1}{q} - \frac{b}{L(t)}\right)g(t). \quad (2)$$

The cutoff terms are bounded and compactly supported away from zero. Therefore $h_f = -tf'$ extends continuously to 0 with value 0 and is bounded with compact support. It can be approximated uniformly by functions $h_{f,n} \in C_c^\infty(0, \infty)$. Formula (1) gives $f_n = (D_+^1)^{-1}h_{f,n}$ in the Schwartz core and $\|f_n - f\|_{1,\infty} = \|h_{f,n} - h_f\|_\infty \rightarrow 0$. Thus f belongs even to the completion interpretation of $\mathcal{T}_\infty^{(1)}(t)$.

Next,

$$\int_0^{1/4} |g(t)|^q dt = \int_0^{1/4} \frac{dt}{tL(t)^{bq}} < \infty, \quad bq = aq + 1 > 1.$$

Equation (2) shows that $h_g = -tg' \in L^q(0, \infty)$. Approximating h_g in L^q by $C_c^\infty(0, \infty)$ and applying (1) proves $g \in \mathcal{T}_q^{(1)}(t)$.

But on $(0, 1/4)$ the choice $b = a + 1/q$ gives

$$|f(t)g(t)|^q = t^{-1}L(t)^{q(a-b)} = \frac{1}{tL(t)}.$$

Hence

$$\int_0^{1/4} |f(t)g(t)|^q dt = \int_0^{1/4} \frac{dt}{t \log(e/t)} = \infty.$$

The source's continuous inclusion $\mathcal{T}_q^{(1)}(t) \hookrightarrow L^q$ therefore implies $fg \notin \mathcal{T}_q^{(1)}(t)$. □

Why the stated integer-order argument misses this

For finite exponents, the usual Leibniz rule and Hölder inequality do indeed give the integer-order claim. At $p = \infty$, however, the proposed norm controls tf' but not f . The first Leibniz term ftg' is thus uncontrolled. The source already flags this defect by noting that $\log(1+t)$ has finite $\|\cdot\|_{1,\infty}$ norm but is unbounded; the construction above localizes the same phenomenon at zero so that it remains inside the Schwartz-core completion.

A precise repair is immediate: if the endpoint norm is strengthened to $\|f\|_\infty + \|f\|_{1,\infty}$, then

$$\begin{aligned}\|fg\|_{1,q} &\leq \|f\|_\infty \|g\|_{1,q} + \|f\|_{1,\infty} \|g\|_{L^q} \\ &\leq C(\|f\|_\infty + \|f\|_{1,\infty}) \|g\|_{1,q}.\end{aligned}$$

Thus the failure is exactly the absent L^∞ control, not an exotic fractional-calculus effect.

Proof intuition and scope

The weight t makes logarithmic singularities cheap: $L(t)^a$ diverges at zero, while $t|(L^a)'| = aL^{a-1} \rightarrow 0$ for $a < 1$. The second factor is placed just inside L^q by a logarithmic power. Multiplication consumes exactly that logarithmic margin and creates the borderline divergent density $dt/(t \log(e/t))$.

This fully refutes Open Question (ii) in its stated exponent range. It does not decide whether a finite-exponent version holds for noninteger α , nor does it address the Stieltjes boundedness/spectrum conjecture (i) or the informal representation conjecture (iii). Exact-ID, title, core-keyword, and official-arXiv searches through 12 August 2026 found no later matching counterexample; this is a bounded novelty audit, not a priority claim.

Reference. P. J. Miana and J. Oliva-Maza, *Generalized Stieltjes and other integral operators on Sobolev–Lebesgue spaces*, arXiv:1906.10772, Remark 3.1 and Open Question (ii).